Analysis of Massive Data Sets

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Prof. dr. sc. Siniša Srbljić

Izv. prof. dr. sc. Marin Šilić

Izv. prof. dr. sc. Goran Delač

Izv. prof. dr. sc. Klemo Vladimir

Faculty of Electrical Engineering and Computing
Consumer Computing Laboratory

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Detection of near-duplicate (textual) documents using the *simhash* algorithm

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Outline

- Motivation
- Fingerprinting
  - Simhash algorithm
  - Rabin's rolling hash function
- IR/NLP basics
- Scalable queries
- Similar methods
  - Random Indexing
  - Image hashing
- Literature

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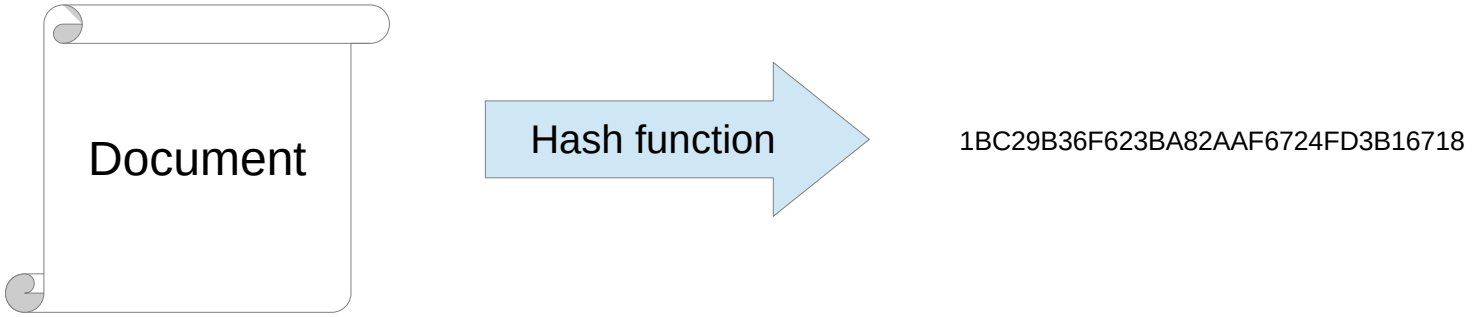
Motivation

- Near-duplicate documents
  - Versioning
    - Different versions of single document
      - Revisions, different file formats, ...
  - Mirroring
    - published in more than one location
  - Plagiarism
    - Exact or “processed” copy
  - Malware
    - Viruses, spam, …
- Scalability
  - Documents and document respositories are large

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Exact copy analysis

- Checksumming
  - Cryptographic hash functions
  - MD5, SHA1, SHA2, ...



  - Catches the smallest edit
    - Great for detection of **exact** copies
    - Not so good for near-duplicate detection
      - Even the smallest change will result in totally different digest

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Near-duplicate detection

- Two families of methods
  - 1. Ranking
    - Information retrieval techniques *1
    - High-dimensional vectors manipulation
  - 2. *Fingerprinting*
    - Document hashing *2
    - Dimensionality reduction

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Fingerprinting

- Similarity preserving hashing
  - X, set of inputs
  - d_x, distance function on X
    - x_1, x_2 elements of X
  - similarity preserving hash function
    - **h: X → Y**
    - |Y| < |X|
  - d_y, distance function on Y

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Fingerprinting

- Similarity preserving hashing
  - Similar inputs have similar hashes

$$\text{if } d_x(x_1, x_2) < \varepsilon_x, \text{ then}$$
$$d_y(h(x_1), h(x_2)) < \varepsilon_y$$

- Illustrative example
  - `h("1234")` = `0xaaaf` `h("5678")` = `0xb115`
  - `h("1235")` = `0xaaae`

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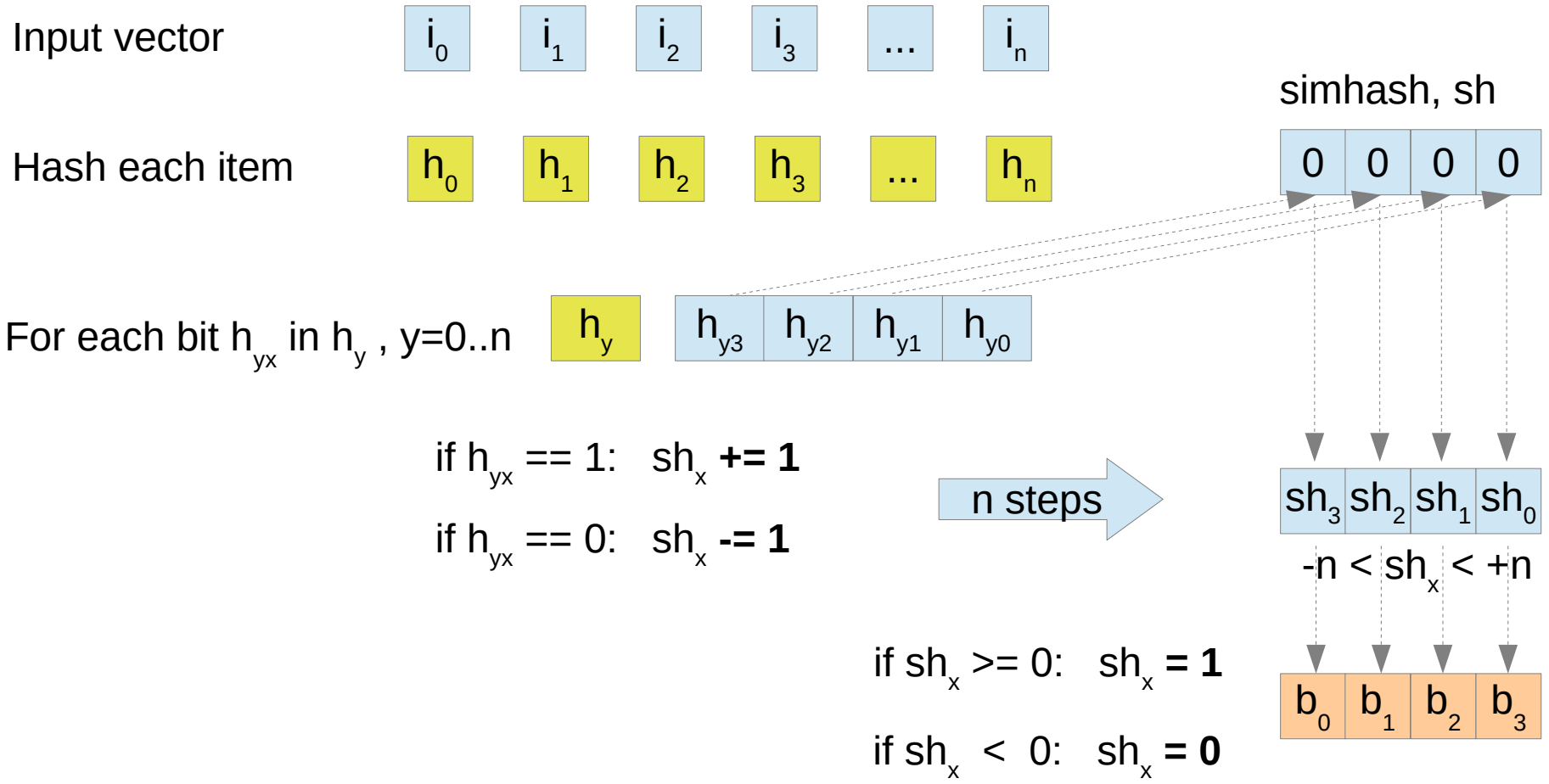
Fingerprinting

- **Simhash** algorithm *3
  - Author: M. Charikar, 2002
  - Fingerprinting technique
    - Fingerprints of near-duplicates differ in a small number of bit positions (**hamming** distance)
  - Dimensionality reduction
    - Maps high-dimensional vectors to small-sized fingerprints (f-bits)
  - Input
    - High-dimensional vector (strings, numbers, …)
  - Output
    - f-bit fingerprint
  - Fingerprint size (f-bits)
    - f is small and arbitrary
    - eg. f=64 for web sites

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Simhash

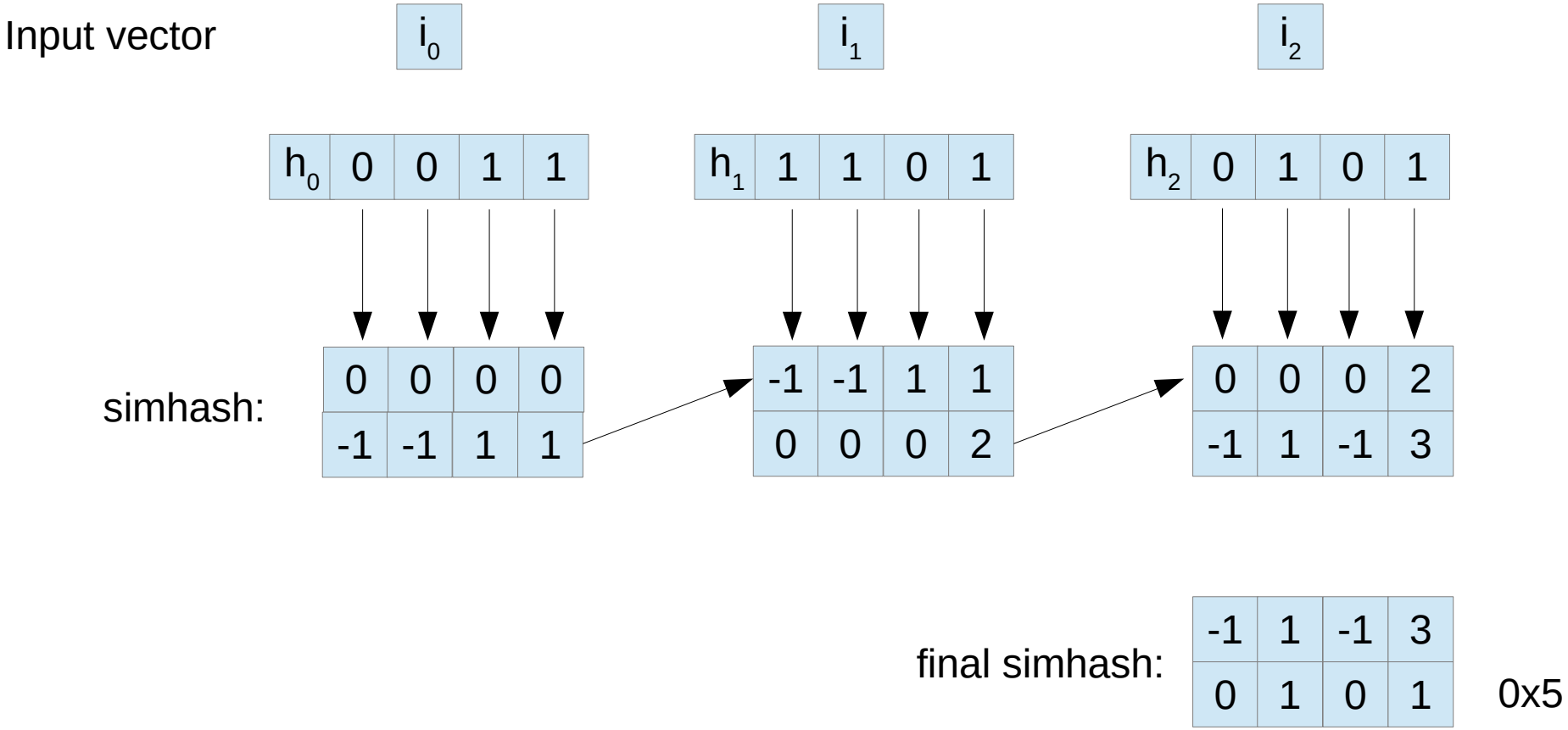
- Simhash computation, f=4



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Simhash

- Simhash example, f=4

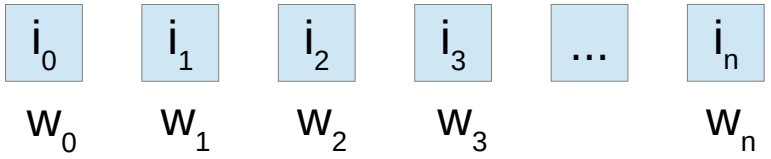


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Simhash

- Weighted Simhash computation
  - Assign weight factor to each feature

Input vector



if h_{yx} == 1: sh_x **+= w_y**

if h_{yx} == 0: sh_x **-= w_y**

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Simhash

- Choice of hash function *h*
  - Uniform distribution
  - Fast
  - Candidates
    - Cryptographic hash functions
      - MD5 (128-bit), SHA-1 (256-bit)
      - Problem: cryptographic hashing is slow
    - Rolling hash functions
      - Input is hashed by moving window element by element

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Simhash

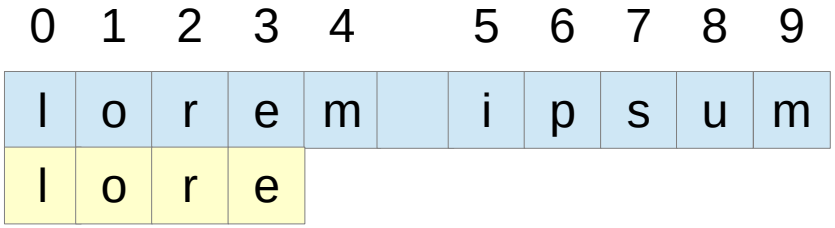
- Choice of hash function *h*
  - Rabin rolling hash
    - Used in Rabin-Karp string searching algorithm
    - Input: string
    - Output: hash - number
  - Computation
    - substring coded as a number with base **d**
      - d = total number of possible characters
    - Coded string at position *i*:

$$x_i = s[i]d^{k-1} + s[i+1]d^{k-2} + \ldots + s[i+k-1]$$

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Simhash

- Choice of hash function *h*
  - Rabin rolling hash
    - Example: k=4, d=32
    - i=0



h(“lore”) = $h(x_o) = x_o \bmod q$ *[q is some large prime number]*

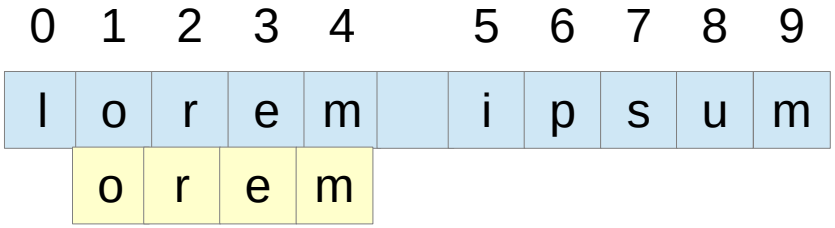
$x_o = int(‘l’)32^3 + int(‘o’)32^2 + int(‘r’)32 + int(‘e’)$

x_{i+1}=?

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Simhash

- Choice of hash function *h*
  - Rabin rolling hash
    - Example: k=4, d=32
    - i=1



h(“orem”) = $h(x_1) = x_1 \bmod q$

$x_1 = (x_0 - \text{int}(\text{‘l’})d^3)d + \text{int}(\text{‘m’})$

$\mathbf{x_{i+1} = (x_i - s[i]d^{k-1})d + s[i+k]}$

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Simhash

- Choice of hash function *h*
  - Rabin rolling hash
    - Fingerprints represented using polynomials
    - Computes hash value of the next string from the previous one
      - Constant number of operations
      - Independent of string length

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IR – A brief excursion

- *How to represent documents? How to calculate simhash weights?*
- **IR – Information retrieval**
  - Models and algorithms for retrieving information from document repositories
- **Statistical NLP – Natural Language Processing**
  - Vector space model
    - Each document is represented as a vector in a high-dimensional space
    - Dimensions are “terms”, values are term frequencies
    - Semantic proximity ↔ Spatial proximity
  - N-gram model for terms definition
    - 1-gram, 2-gram, 3-gram, ...

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IR – A brief excursion

- N-gram model
  - Tokenization examples

“lorem ipsum dolor sit amet”

1-gram word

“lorem”, “ipsum”, “dolor”, “sit”, “amet”

2-gram (bigram) words

“lorem ipsum”, “ipsum dolor”, “dolor sit”, “sit amet”

3-gram (trigram) characters

“lor”, “ore”, “rem”, “em ”, “m i”, …

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IR – A brief excursion

- N-gram model
  - Document collection is usually represented in a matrix form
    - Rows – documents
    - Columns – terms (1-word)

```
D1 = “lorem ipsum dolor sit lorem amet sit”
D2 = “lorem ipsum adipiscing ipsum elit amet ipsum”
```

	adipiscing	amet	dolor	elit	ipsum	lorem	sit
D1	0	1	1	0	1	2	2
D2	1	1	0	1	3	1	0

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IR – A brief excursion

- Calculating similarity between documents
  - Document similarity = vector similarity!
  - Usually, ***cosine*** similarity is calculated
- Adjusting term weights (**tf-idf** model)
  - $TF_{i,j}$ = term frequency of w_i in document D_j
  - Some *tf* schemes:
    - $tf(TF_{i,}) = TF_i$
    - $tf(TF_{i,}) = log(1 + TF_{i,})$
  - Problem - terms that are rare vs. terms that are common
    - N – total number of documents
    - DF_i = number of documents that contain term w_i
    - $idf_i(N, DF_i) = log(1 + N/DF_i)$
  - Final tf-idf scheme
    - $tf\text{-}idf(TF_{i,,} N, DF_i) = tf(TF_{i,}) * idf(N, DF_i)$

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IR – A brief excursion

- N-gram model
  - Number of dimensions = number of terms
    - Small N: dissimilar documents appear similar
    - Large N: similar documents appear dissimilar
  - Example:
    - Vocabulary size = 20000
    - 2-gram dimensionality ~ 20k x 20k = 400 mil.
    - 3-gram dimensionality ~ $8 * 10^{12}$
  - Reducing dimensionality
    - Removing “stopwords” - words that are common in every document (e.g. “the”, “it”, “an”, …)
    - Stemming – reducing words to a stem (base or root form)
      - Problems: two different words can have equal stem and stems can be unintelligable to users

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Fast Queries

- **F**, collection of f-bit fingerprints
- **Q**, query
  - single or set of fingerprints

- Task
  - identify whether **Q** differs from any of the fingerprints in ***F*** in at most **k** bits

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Fast Queries

- Example numbers
  - 8B 64-bit fingerprints = 64GB
  - Online query
    - Q = single fingerprint
    - Restriction: few milliseconds
  - Batch query
    - Q = set of fingerprints
      - e.g. |Q| = 1M
    - Restriction: ~100seconds
      - 1B queries per day

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Fast Queries

- 1. approach
  - Build sorted table of **F**
  - Build list **Q'** with all fingerprints whose Hamming distance from **Q** is at most **k**

Q
100

k=1

Q'
1 0 0
1 0 1
1 1 0
0 0 0

F
0 0 0
0 0 1
0 1 0
0 1 1
1 0 0
1 0 1
1 1 0
1 1 1

  - 64-bit fingerprints (k=3)
    - $|Q'| = \binom{64}{3} = 41664$

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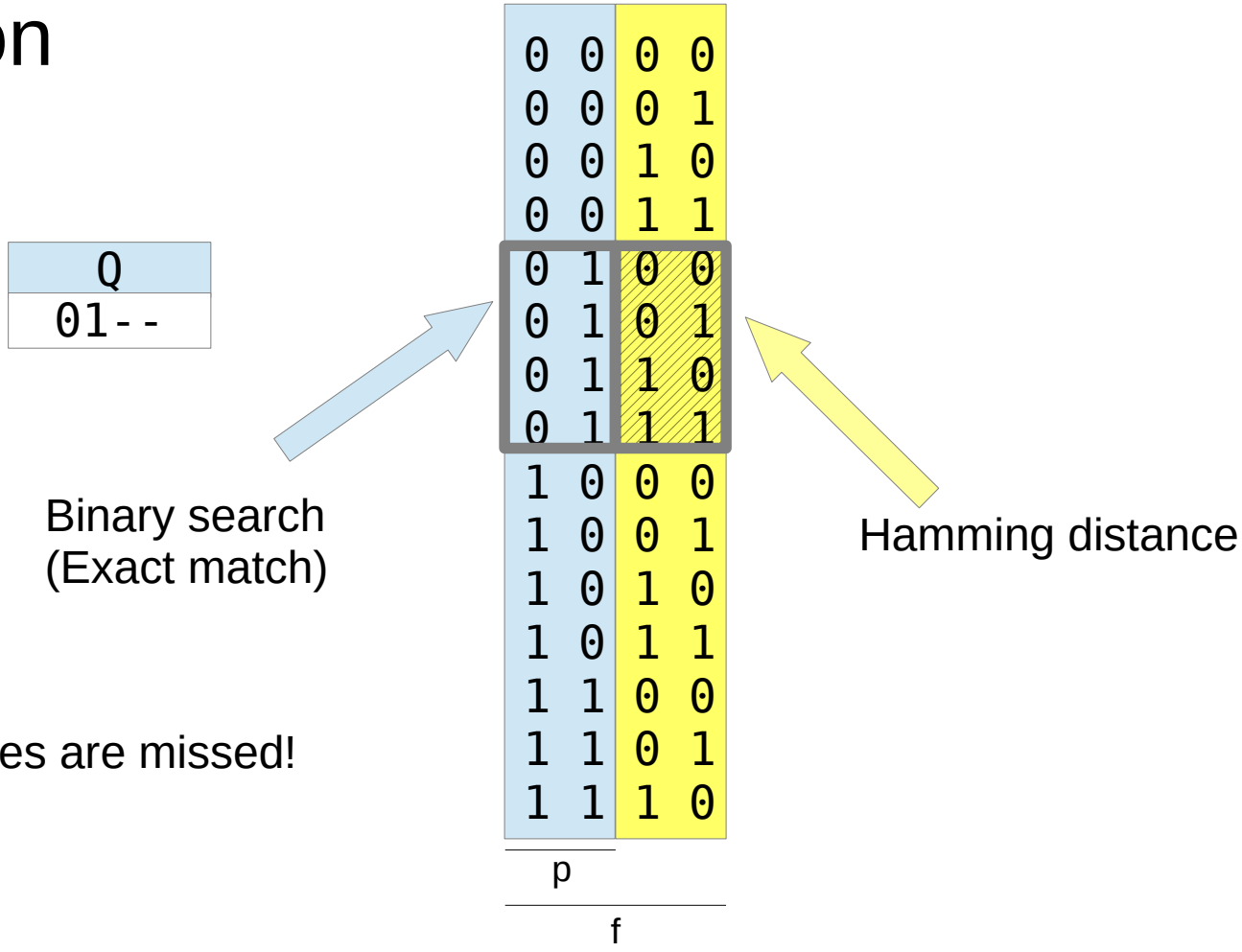
Fast Queries

- 2. approach
  - 1. Build sorted table of **F**
  - 2. Find set of fingerprints (**F'**) that have equal most significant part (p bits)
    - Sorted table – binary search O(p)
  - 3. Check Hamming distance for each fingerprint in **F'**
  - This approach will locate all fingerprints in **F** that differ in at most k bits
    - **BUT, restricted to (only) least significant f-p bits!**

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Fast Queries

- Illustration

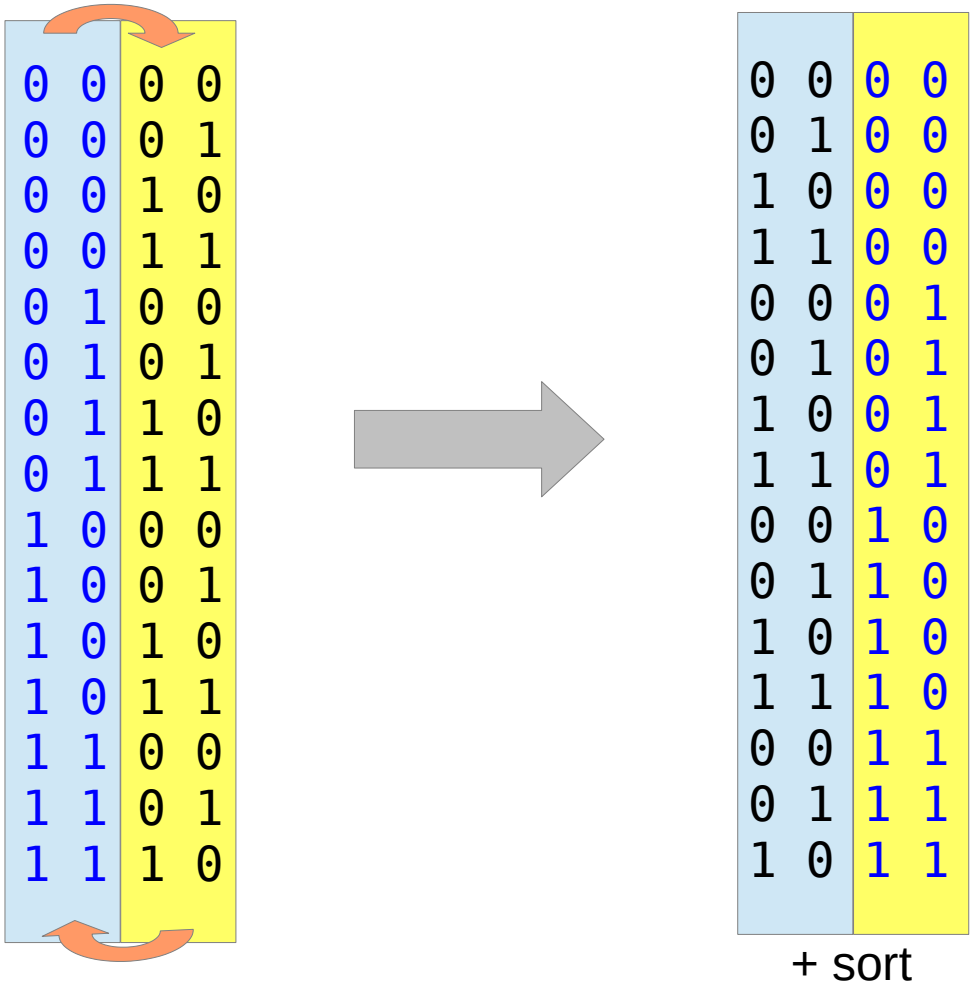


Since p < f:
some true positives are missed!

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Fast Queries

- Increasing precision/recall
  - Generate additional table
    - Reversed positions



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Fast Queries

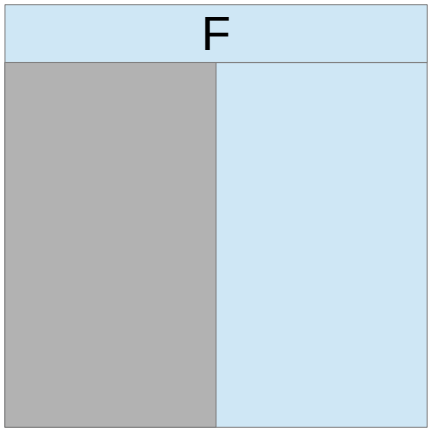
- Solution
  - Build additional tables
    - Each with *different permutation* of bits
    - Every table has different set of significant bits

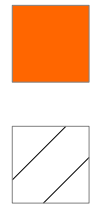
- Algorithm for fast (online) queries
  - Build **t** sorted tables of fingerprints: $T_1, T_2, \ldots, T_t$
  - Each table T_i also contains
    - p_i – number of significant bits
    - Π_i – random permutation
  - *Every fingerprint in T_i is permuted with permutation Π_i*

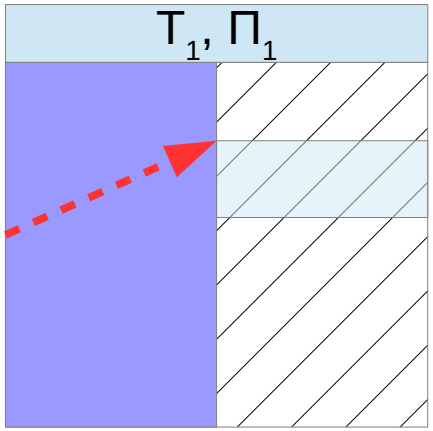
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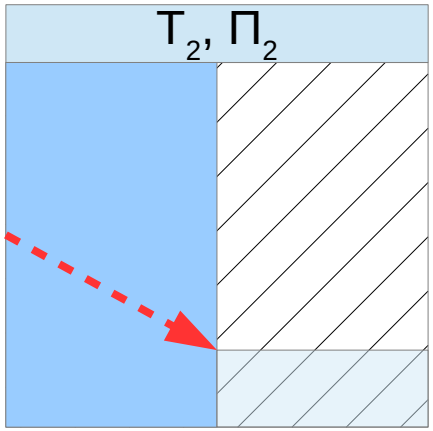
Fast Queries

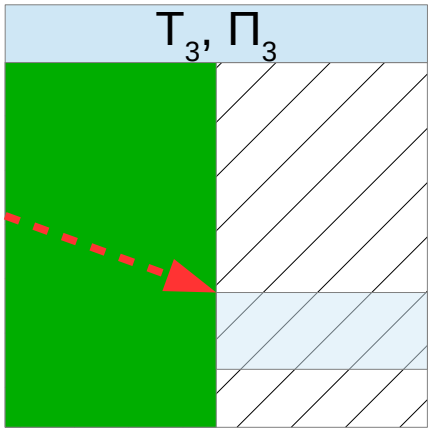
- Illustration
  - Build tables
  - Query in parallel

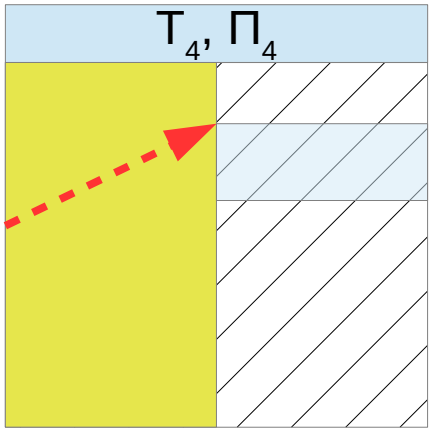












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Fast Queries

- For given **Q** and **k**
  - Read each table (in parallel)
    - 1. Get fingerprints in T_i whose significant p_i bits match the significant p_i bits of $\Pi_i(Q)$
      - T'_i
      - $O(p_i)$ steps
    - 2. For each fingerprint in T'_i, check if it's Hamming distance is at most k bits from $\Pi_i(Q)$

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Fast Queries

- Example with t=20, f=64, k=3, |F| = 8B (2^{34})
  - Split f into 6 blocks (4x11 + 2x10 bits)
  - Select 3 out of 6 blocks $\binom{6}{3}$ = 20 ways
  - Arrange those blocks as significant bits
  - p = sum of those bits
    - 31, 32, or 33
  - On average query returns 2^{34-31}=8 fingerprints

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Fast Queries

- t and p parameters
  - t ~ p
  - Query time ~ 1/p
  - Storage requirements ~ p
  - Space/time tradeoff
    - Analytical solution for t *2

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Experimental results

- Detecting near-duplicate web pages
  - Web-crawling in Google
- Database
  - 8B fingerprints, k=1..10
- Manually tag experimental data set
  - True/false positive/negative
- Precision/recall graph
  - Precision: #tp / # returned results
  - Recall: #tp / # expected results

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Experimental results

- Precision
  - #tp / # returned results
  - Percentage of relevant items in the returned set
  - Example
    - Qr = [**d3**, **d1**, d4, **d2**, d5], $relevant_{Qr}$ = {d1, d2, d3}
    - P = 3/5=60%
- Recall
  - #tp / # expected results
  - Percentage of all relevant documents in the collection that is in the returned set
  - Example:
    - Qr = [**d3**, **d1**, d4, **d2**, d5], $relevant_{collection}$ = {d1, d2, d3, d7, d8, d9, d10}
    - R = 3/7=43%

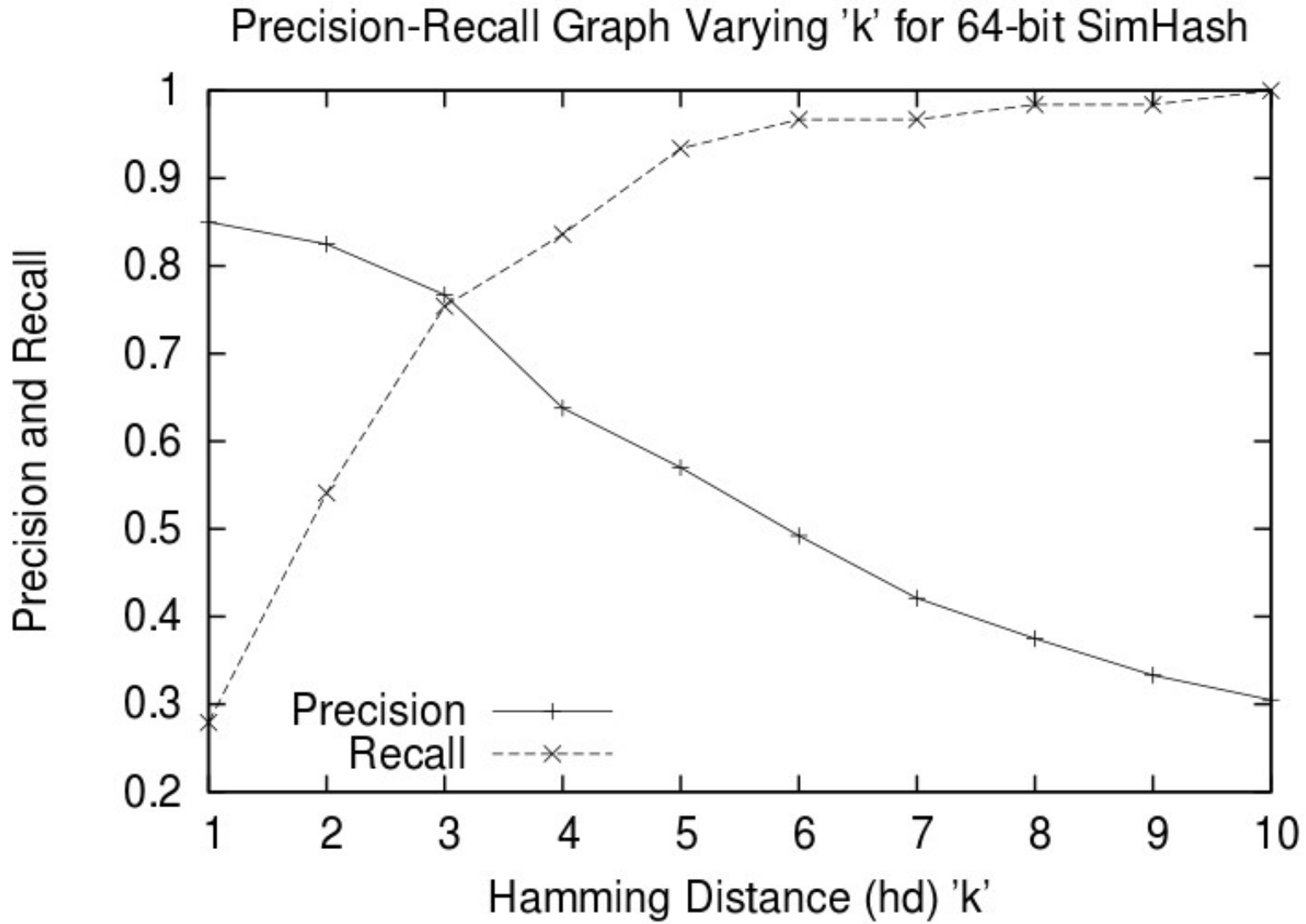
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Experimental results

- Two extreme cases:
  - Qr is whole collection
    - Recall is 100%
    - Precision is low
  - Qr is small
    - Precision is high
    - Recall is low
- F-score
  - Combination of Precision and Recall
  - $F_1 = 2PR / (P + R)$ (harmonic mean)

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Experimental results



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Similar methods

- Random Projection / Random Indexing *5
  - Dimentionality reduction technique
  - Input
    - d-dimenional data
  - Output
    - k-dimensional data
    - k << d
  - Procedure
    - Construct a random d *x k* matrix *R*
    - Project d-dim input data to a smaller k-dim space by multiplying input data with matrix *R*

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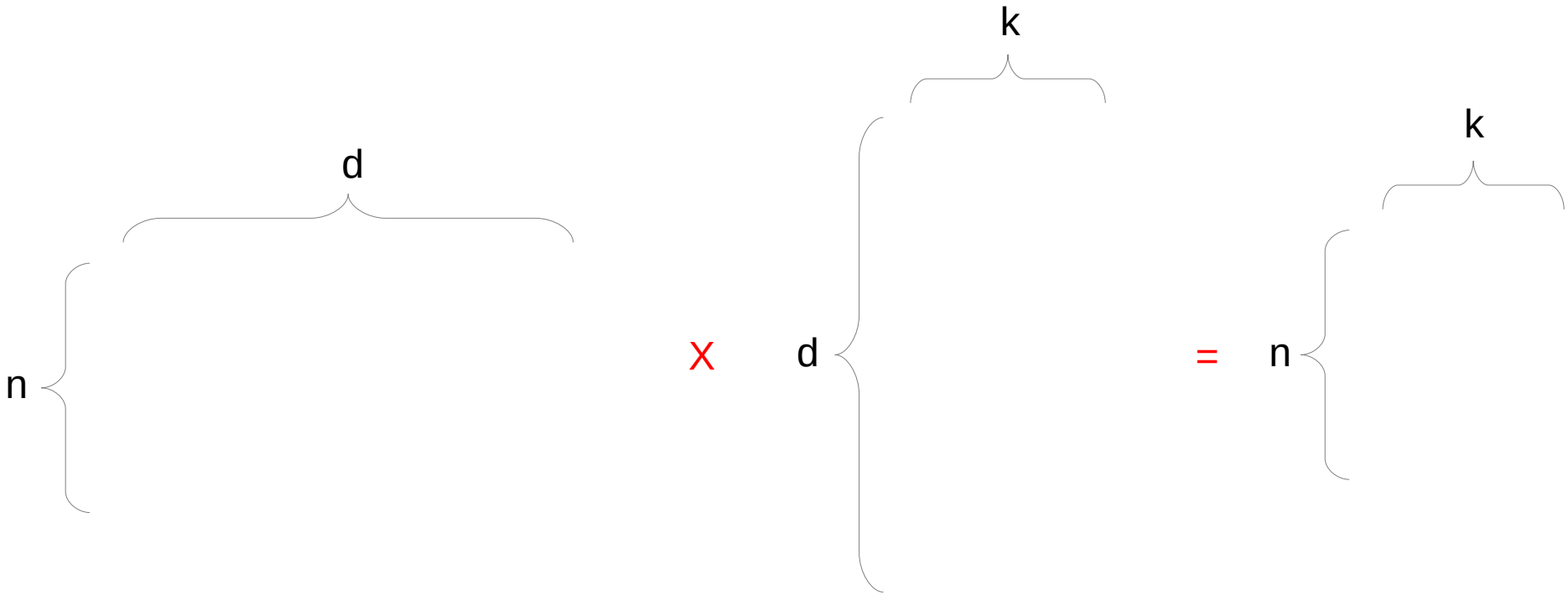
Similar methods

- Random Projection / Random Indexing
  - Random d *x k* matrix *R*
    - *Johnson-Lindenstrauss Lemma*
      - If points in a vector space are projected onto a randomly selected subspace, then the distances between the points are approximately preserved
    - Ideally, *R* should be orthogonal
      - For high-dimensional data, random directional vectors are *good enough*
    - *R* is usually constructed as sparse Gaussian distributed random matrix

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Similar methods

- Random Projection / Random Indexing



n = # of input documents
d = # of features
k = # of dim. reduced features

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Similar methods...

- PCA (Principal Component Analysis)
- LDA (Linear Discriminant Analysis)
- MF (Matrix Factorization)
  - Stohastic Gradient Descent (SGD)
- Autoencoder
  - usually as neural models
- Data sketching
  - Data streams lecture

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Literature

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